

# CareTrace

A provenance-first longitudinal care note that compresses fragmented interactions for action while keeping every AI-supported highlight anchored to immutable evidence.

|                    |                            |                                 |
|--------------------|----------------------------|---------------------------------|
| React / Vite       | REST / JSON + signed token | FastAPI + RBAC                  |
| Fast UI            | Clinic scope on every call | Service + validation            |
| Precomputed Glance | SQLAlchemy                 | SQLite + field-level encryption |

RAW

→ REDACT

→ CANDIDATE

→ MATCH

→ CITE

→ GLANCE

## Why this shape

Healthcare collaboration needs compression, but trust fails when a summary cannot show where it came from. CareTrace therefore treats AI output as a candidate, not a fact. The backend validates evidence, owns offsets, stores provenance, and may return abstained. Page loads read a precomputed Glance snapshot and never wait for an LLM.

### Trust boundary

Raw interactions are committed and encrypted before AI work. Known names, phones and IC/ID patterns are redacted and checked before a provider call. Provider failures preserve the source and previous Glance. Logs and audit events contain metadata only.

### Provider boundary

Ollama chat and embedding calls sit behind explicit provider interfaces. AI\_PROVIDER=ollama|fixture is selected at startup and never changes silently. The fixture provider makes automated tests and offline evaluation reproducible.

Synthetic data only. CareTrace is priority support, not diagnosis or clinical decision support.

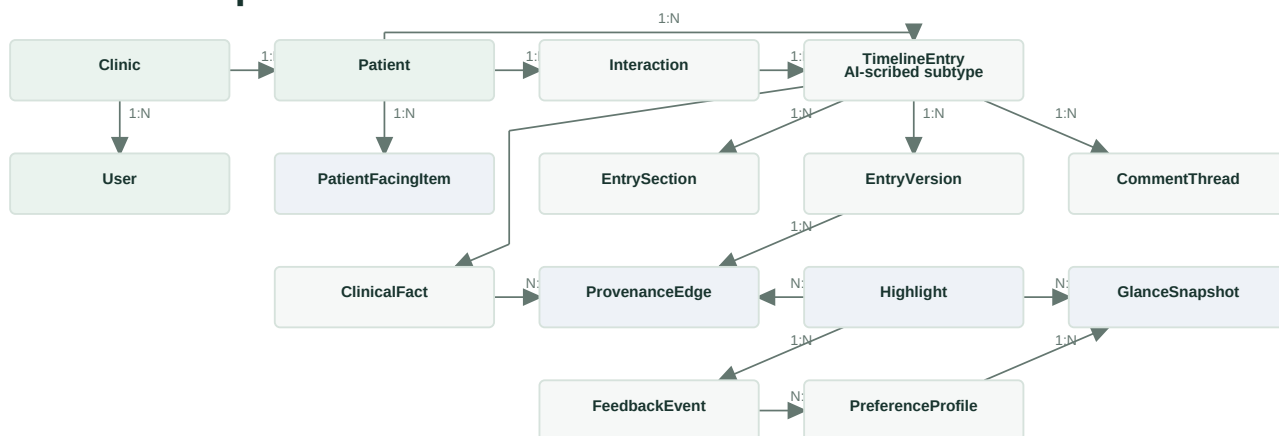
# Evidence before emphasis

## Backend-owned provenance

The LLM returns `source_ref`, a verbatim `evidence_quote` and a normalized candidate - never character offsets. The backend searches only the referenced immutable source version. Trust outcomes are verified (unique exact quote), supported (unique normalized match), review\_required (ambiguous, missing, malformed or wrong source), and abstained (no eligible claim). Only verified and supported can enter Glance.

Redaction creates a boundary map from redacted positions to the encrypted original. The stored `ProvenanceEdge` contains source entry/version, section, both coordinate systems, match method and a quote hash. Every read revalidates the hash. Later edits create new versions, so citations do not drift.

## Relational spine



Arrows follow stored FK/reference direction. `TimelineEntry` includes AI-scribed-note subtypes; `GlanceSnapshot` stores ranked `Highlight` IDs.

## Deterministic collaboration

Staff and clinicians own separate sections. Independent section counters allow simultaneous edits without overwriting each other; a stale same-section write returns 409. Revert restores one authorized section into a new version. Conflicting allergy or dosage facts enter a review queue; only a clinician can select the authoritative fact, and both sources remain.

## Bounded adaptive ranking

**final = max(applicable risk floor, clamp(rule score + learned bonus, 0, 100))**

Risk floors are intentionally narrow: explicit synthetic risk tag, allergy conflict, dosage conflict, critical unresolved action or clinician-confirmed warning. Generic symptoms never create a floor. Clinical entities affect ordering only, not diagnosis.

Explicit positive and negative feedback updates per-user embedding centroids. The similarity bonus is clamped to +/-8; non-interaction is ignored; critical floors cannot be suppressed.

# Know when the system is wrong

| verified                            | supported                                | review_required                                           | abstained                                   |
|-------------------------------------|------------------------------------------|-----------------------------------------------------------|---------------------------------------------|
| Unique exact quote<br>Auto-eligible | Unique normalized match<br>Auto-eligible | Ambiguous / missing / malformed<br>No automatic highlight | No eligible claim<br>No automatic highlight |

## Security and patient boundary

Passwords use scrypt and tokens use HMAC-SHA256. Every query rechecks subject, role and clinic against SQLite. SQLite uses field-level encryption for raw interactions, redaction maps and patient names; the database file is not claimed to be encrypted. Patients may submit their own patient\_input through POST /entries but cannot read the internal timeline. Only clinician-approved PatientFacingItem rows are visible to them. Local HTTP is for demonstration only; deployment requires TLS termination and managed secrets.

## Automated evaluation

Four required micro-test modules cover RBAC, revision history, highlight provenance and concurrent edits. The separate bonus test covers self-learning importance and its bounded learned bonus. Additional tests verify review\_required and abstained outcomes, PHI blocking, the risk floor boundary and critical-item survival after rejection.

|         |         |          |
|---------|---------|----------|
| 1.16 ms | 1.44 ms | < 300 ms |
| median  | P95     | target   |

Warm-path approximation: local in-process HTTP client, 500 timeline entries, 20 warm-up requests and 500 measured GET /glance requests. This isolates application/database time and excludes proxy, network and production concurrency effects.

## Scope and migration

Configurable prototype defaults are DATA\_DECAY\_THRESHOLD\_DAYS=90, RECENT\_CONTEXT\_COUNT=3, RAG\_TOP\_K=5 and GLANCE\_MAX\_ITEMS=5. Voice, cloud deployment, real EHR integration and formal compliance are out of scope. A production migration would add a durable queue, managed database, key rotation, operational telemetry, extraction datasets and reviewed clinical governance before real patient use.

|                  |                                                                                            |
|------------------|--------------------------------------------------------------------------------------------|
| DESIGN PRINCIPLE | Compression may be probabilistic. Provenance, permissions and overrides are deterministic. |
|------------------|--------------------------------------------------------------------------------------------|